STIVEN CATALYST

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# MANAGEMENT REVIEW

Management without theatre.




OPERATIONS

## STANDARD WORK AND KAIZEN

A wartime pocket card, what the plants reported, Imai's two parts of management, the three elements of standard work, ten years after the training, Toyota's 55 seconds, and a card for one kaizen.

**55** seconds for the seven tasks of fitting a car seat at Toyota: a written standard that shows a gap as it happens.

SPEAR & BOWEN, 1999

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NO. 35 · OPERATIONS

# In this issue

A standard that nobody changes goes stale, and an improvement that nobody writes down is lost by the next shift. This issue is about the link between the two: the written best way to do a job today, and the habit of finding a better one.

In the Second World War, the US Training Within Industry service taught supervisors to teach a job the same way and to improve it, with four steps on a pocket card; Toyota later adopted the programmes. In 1986 Masaaki Imai brought the word kaizen to Western managers: improvement involving everyone, built on standards that are kept. A study of 11,575 firms found the wartime training still showing in sales and productivity ten years later, and at Toyota a seat is fitted in a written sequence timed in seconds.

*Stiven Janaqi* EDITOR · STIVEN CATALYST


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## HOW TO READ THIS ISSUE

**Figure**
Every figure has its source and year at the foot of its page.

**Our reading**
Where the editors interpret rather than the research, it says so.

**Practice**
The steps and the card are proposals to try, not research results.

COVER STORY

# Until a better way is found

From 1940 to 1945, the US government's Training Within Industry service (TWI) trained supervisors in war plants. Each of its methods was so simple that it was printed on a pocket card.

**01**

**Job Instruction**

how to teach a job: important steps and key points, then show, try out, follow up

**02**

**Job Methods**

how to improve a job: break it down, question every detail, develop and apply the new method

**03**

**Job Relations**

how to handle a problem with people: get the facts, weigh and decide, act, check the results

The Job Methods card ends with one line: put the new method to work and use it until a better way is developed. By September 1945 TWI counted 1,750,650 certificates in 16,511 plants and unions. Toyota later adopted the programmes; Job Instruction is still the main training tool of its team leaders, says the Lean Enterprise Institute.

OUR READING

Standard work and kaizen fit on one card: the new method becomes the standard, and the standard waits for the next better way.

Sources: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945; Lean Enterprise Institute, 2014

The counts are TWI's own. That Toyota adopted TWI comes from the Lean Enterprise Institute; the sources we checked do not give the year.

THE NUMBERS

# What the plants reported

Asked by the US House of Representatives in 1943 for its results, TWI tabulated the voluntary reports of plants seven times. The last, September 1945:

PLANTS REPORTING A RESULT OF 25% OR MORE, BY KIND OF RESULT, SEPTEMBER 1945




In May 1943 most results were still under 25%. In one aircraft plant, a department where every supervisor held the Job Instruction certificate had a scrap cost of 15 cents per head; the plant average was 61.

OUR READING

Plants chose whether to report, and TWI used the reports to back its budget requests. They show what managers saw; the comparison is on page 6.

Source: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945

Each share is of the plants that reported that kind of result; the report does not say how many plants each tabulation counted. The first one used more than 600 plant statements.

THE MODEL

# Maintain and improve

In Kaizen (1986), Masaaki Imai defines kaizen as ongoing improvement involving everyone, managers and workers alike. In his account management has two parts: maintaining the current standards, through training and discipline, and improving them.

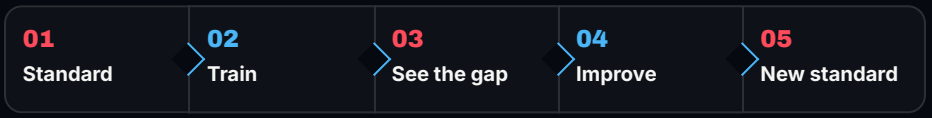
**01**
**Takt time**
the pace at which products must be made to meet customer demand

**02**
**Work sequence**
the order in which one operator does the tasks within the takt time

**03**
**Standard inventory**
the stock inside the process, including units in machines, that keeps it running smoothly

These are the three elements of standardized work for the Lean Enterprise Institute. Posted at the workstation, it is the baseline for the next kaizen. Its lexicon quotes Taiichi Ohno: there can be no kaizen without a standard.

THE CYCLE, AS THE EDITORS DRAW IT




OUR READING

A standard nobody follows is only a document. An improvement that never becomes the standard lasts as long as the person who made it.

Sources: Masaaki Imai, McGraw-Hill, 1986 (via Quality America, Kaizen); Lean Enterprise Institute, Lean Lexicon: Standardized Work, 2014; Lean Enterprise Institute, Lean Lexicon: Kaizen, 2014

Imai's definition and his two parts are confirmed in summaries of the book, not in the book itself.

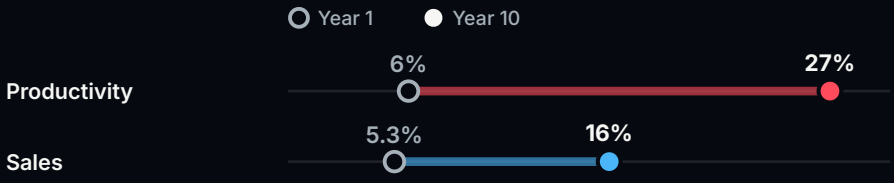
More in the essay A good SOP is not a document

WHAT THE RESEARCH SAYS

# Ten years after the training

Nicola Bianchi and Michela Giorcelli collected data on all 11,575 US firms that applied to TWI. Money and trainers ran short: 48% received no training, and only 7% received all three modules.

TRAINED AGAINST UNTRAINED APPLICANTS, ONE AND TEN YEARS AFTER THE TRAINING




Before the training, both groups followed the same trends. Sales of trained firms peaked in the eighth year, 21.7% above the others. Firms trained in Job Instruction began to set standard procedures for operations, and the gains held even as many top managers left after the war.

OUR READING

The training lasted because it changed how the work was organised, not only what one manager knew.

Source: Nicola Bianchi & Michela Giorcelli, Journal of Political Economy, 2022

A natural experiment, not a randomised trial. The figures are those of the article in the Journal of Political Economy, 2022.

HOW IT IS MEASURED

# A gap you see in seconds

Steven Spear and Kent Bowen studied more than 40 plants over four years. Their first rule of the Toyota Production System: all work is highly specified as to content, sequence, timing and outcome.

At Toyota's plant in Georgetown, Kentucky, fitting the right-front seat of a Camry was designed as seven tasks in 55 seconds. A worker still on task 4 after 40 seconds, when it should end at 31, is behind; the floor of the zone is marked in tenths, so he and his team leader see it at once.

| | | |
|---|---|---|
| 01 | **Not done as specified** | Find the true skill of the person or capability of the machine; train or modify. |
| 02 | **Done as specified, still defective** | Change the design of the work: the standard itself was wrong. |

HYPOTHETICAL EXAMPLE, A PACKING STATION

| | |
|---|---|
| **Standard** | six steps in 90 seconds; the label is checked at step 5 |
| **Observed** | step 5 skipped on 4 of 20 orders at the evening peak |
| **Response** | the scanner sits too far away: move it, test for a week, rewrite the standard |

A gap in the standard, not in the person. The case is invented.

Source: Steven Spear & H. Kent Bowen, Harvard Business Review, 1999

The Camry example dates from 1999. The two responses follow Spear and Bowen; the packing example is the editors'.

TOOL OF THE ISSUE

# The kaizen card

One card for one improvement to one standard. Fill it in where the work is done, with the people who do it; when the cause needs digging, the 5 Whys worksheet helps.

**01 The standard** the job, its important steps and key points, the date of the version

**02 Time** what the standard allows, what it took today

**03 The gap** what departed from the standard, where, how often

**04 Not followed, or wrong?** train the person, or change the standard

**05 One small change** who tries it, until when, measured the same way as before

**06 New standard** written, posted, taught with the four steps of Job Instruction

Sources: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945; Steven Spear & H. Kent Bowen, Harvard Business Review, 1999; Masaaki Imai, McGraw-Hill, 1986 (via Quality America, Kaizen)

A practice proposed by the editors, after TWI's Job Instruction and Job Methods cards (1945), Spear & Bowen (1999) and Imai (1986).

Open the tool: 5 Whys stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

# Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.


**The Training Within Industry Report, 1940-1945**
War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945
https://www.govinfo.gov/content/pkg/GOVPUB-PR32_5200-00275cf1fd80a1d51995ae716fd38f34/html/GOVPUB-PR32_5200-00275cf1fd80a1d51995ae716fd38f34.htm

**Lean Lexicon: Training Within Industry (TWI)**
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/training-within-industry-twi/

**Kaizen: The Key to Japan's Competitive Success**
Masaaki Imai, McGraw-Hill, 1986 · via Quality America, Kaizen
https://qualityamerica.com/LSS-Knowledge-Center/qualityimprovementtools/kaizen.php

**Lean Lexicon: Standardized Work**
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/standardized-work/

**Lean Lexicon: Kaizen**
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/kaizen/

**The Dynamics and Spillovers of Management Interventions: Evidence from the Training Within Industry Program**
Nicola Bianchi & Michela Giorcelli, Journal of Political Economy, 2022
https://www.nber.org/papers/w28833

**Decoding the DNA of the Toyota Production System**
Steven Spear & H. Kent Bowen, Harvard Business Review, 1999
https://hbr.org/1999/09/decoding-the-dna-of-the-toyota-production-system


**EDITORIAL METHOD**

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



# MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

## ALL ISSUES

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